Research Templates

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Contents

Author Response Template

One section per cluster, four sentence roles per section

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1 Is a template encouraged practice

Question 1 asks this directly, and records a suspicion alongside it: that venues are deliberately limiting what a response can contain, and that they may want only the factual problem corrected.

The suspicion about scope is right. The practice is encouraged, but for a document that is not this one.

Templating a response is ordinary and the filled-in artifacts are everywhere: community LaTeX templates, a Typst package, lab templates, a publisher's fillable form. Every one of them is a journal response letter. They quote each reviewer comment, reply underneath it, cite the page and line where the manuscript changed, and compile to a document that travels beside a revised manuscript. No program committee publishes one. The strongest statement from a venue body is SIGPLAN's guidance to its chairs, which encourages holding a response phase at all and says nothing about the shape of the response.

So the format is supplied for the wrong genre. A 750 word plain text box, one box for three reviewers, no revised manuscript to point into, and no line numbers, cannot carry a per-comment reply at all.

On the limiting, the call for papers of the IEEE Symposium on Security and Privacy for 2027 says it outright. The rebuttal "should exclusively focus on factual errors in the reviews and concrete questions posed by the reviewers," and "New research results or other additional material cannot be provided." The same page adds that failure to follow the instructions "will result in immediate rejection." That is the scope, written by the people who set the limit. It is not a hint to be inferred from the word count.

What follows for this template. Scope is constrained and depth is not. The venues limit what a response may be about. None of them says how deeply the things in scope may be answered. A shallow answer therefore recovers nothing that a deep one would have cost.

2 Do people cluster as corrections and discussions

Question 3 asks whether that split is a recognized one and whether templates for it exist.

It is not recognized. DISAPERE annotates 20,000 sentences from 506 review and response pairs at ICLR 2019 and 2020, and its scheme has fourteen response actions in three stances. Nothing in it splits corrections from discussions, and no venue's guidance uses that division either. There is nothing to copy, so there is no cost to inventing a structure that suits the panel in front of you.

Three of its numbers are worth keeping. Answering a question is 32.76 percent of all response sentences, three times the next argumentative label. A response is mostly answers. Rejecting a criticism is 10.37 percent against 2.70 for conceding one, so refusing a reviewer, with a reason, is the common case rather than the risky one. Claiming a task is already done is 8.58 percent against 2.01 for promising it by the deadline, and that is at a venue with far more room to promise than this one.

One number cuts the other way. Contradicting something the review presented as fact is the rarest argumentative label in the set, 0.86 percent. The security venues invite that move ahead of every other. The thing most wanted here is the thing least done where the counting was possible.

So the clustering rule in this template is not corrections against discussions. It is root cause, which is the rule already written into the procedure on the site: the root of a comment is the thing that would have to change for the comment to go away, and comments sharing a root are answered once, as a group, with the group named.

3 The macros

Paste these into the response preamble. Every one of them is stripped by the exporter, so the working apparatus never reaches the submission form.

```
% One cluster. #1 the kind of engagement, #2 the subject.
\newcommand{\rcluster}[2]{\subsection*{#1: #2}}

% The four sentence roles, as reminders while drafting. Delete before export.
\newcommand{\ranswer}[1]{#1}     % the direct answer, first
\newcommand{\rground}[1]{#1}     % the fact, number, equation, or passage
\newcommand{\rconseq}[1]{#1}     % what follows, formally and operationally
\newcommand{\rchange}[1]{#1}     % what the revision does, checkable

% The reviewer's own phrase, echoed in the first clause. Marks it so the
% coverage tool can confirm every cluster carries one.
\newcommand{\recho}[1]{#1}

% A commitment that passed the gate. #1 configuration, #2 table it lands in.
\newcommand{\rcommit}[2]{#1, reported in #2}

% An offer, numbered, with what it would show. Never framed as required.
\newcommand{\roffer}[2]{#1, which would show #2}
```

4 The skeleton, one cluster

```
\rcluster{Kind}{Subject}
\recho{[the reviewer's own distinctive phrase]}:  \ranswer{[the direct answer, as
asked]} \rground{[the fact, number, or passage that supports it]} \rconseq{[what it
means formally, and what it means for whoever would use the system]} \rchange{[what
the revision does, stated so a shepherd can check completion]}
```

Kind, chosen before drafting. A clarification of. A consideration of. A misunderstanding about. The hidden assumptions of. The motivation for. A title naming only a topic tells the reader nothing about what the section does with it.

The echo is the first clause, not a separate sentence. It costs about ten words per paragraph and it is the last thing to cut. Without it a correct sentence answers a question the reader has not been reminded of.

The four roles are an order, not a length. Any of the four may be one clause. None may be absent, except that a cluster with nothing to change omits the change rather than inventing one.

5 Posture, selected by the classification

The classification step assigns every comment one of four kinds, and the kind decides the posture. This table is the mapping.

Kind	Premise	Posture
Misreading	The manuscript contradicts it	Observe that there appears to be a misunderstanding, then the correct fact. Never a statement that the reviewer is wrong. Treat it as evidence the text was unclear.
Defect	The manuscript confirms it	Agree first. Thank them where they spelled out the fix. Then the fix, without dwelling on how the error arose.
Scope question	Asks where the claim's boundary sits	Name the boundary. If the case sits outside the threat model, say so and say where the boundary is.
Request	Asks for work	Through the gate. Commit, decline with an honest reason, or offer.

The declining rule, both halves in one paragraph, always. Refute the phrasing where it is inaccurate, and honor the concern underneath it. A refusal without the second half reads as defensive. A concession without the first gives up ground the reviewer did not ask for.

6 The gate, before any commitment enters a draft

Three questions in order. No commitment passes without the author's sign-off.

1. Was the original choice justified, and is the justification written where the reviewer looked? If so, the answer is that justification and no work is promised.
2. Is the work feasible inside the response window or the shepherded revision? If not, decline with an honest reason.
3. Does it require advancement at the scale of a follow-up project? If so it is not promised, because promising work at that scale concedes the paper is a major revision at best.

Two constraints on what leaves the gate. A promise names a configuration and a table, because a shepherd can hold nothing else. A decline states a reason that is never effort and never the word limit.

Work you are unsure about becomes a numbered offer with what it would show, and the paper must not appear to depend on the reviewers accepting one.

7 Filling it

The order of the work is the nine step procedure on the website at `resources/research/respond/procedure.md`. It is not repeated here. The two steps that touch this file are step seven, which drafts one section per cluster, and step eight, which runs the wording pass and the per-question audit.

Only step seven writes. Everything before it extracts, reads, clusters, classifies, and gates.

Step seven expanded slot by slot, with the word budget, the refuse-to-start list and the exit tests, is `template_rebuttal_procedure.tex` beside this file.

8 The cut order, when it runs long

Three places, in this order, and none of them costs anything.

Restatement of the manuscript first. The reviewers have read the paper.

Spectator voice second. State your own numbers rather than reconstructing them from published figures in front of the reader.

Pointing instead of answering third. A section number is not an answer, since the reviewer already read that section, which is how they formed the comment.

If it still runs long, cut the section answering the least movable review.

9 What never reaches the form

Comment identifiers. Status markers. Approval labels. Notes on open decisions. Gates addressed to the author. Any working apparatus at all. The plain text is generated from this source rather than kept as a second copy by hand, because a second copy drifts and the drift is found at the form.

10 Sources

- The procedure, the drafting shape, the postures, the phrasing shelf, and the cut order are all the author's own, written at `content/resources/research/respond/` on the website. This template operationalizes them and adds nothing to them.
- The scope sentences quoted in section 1 are from `sp2027.ieee-security.org/cfpapers.html`, the call for papers of the IEEE Symposium on Security and Privacy for 2027, read from the page.
- The survey of what response templates exist covers the Overleaf community templates tagged for responses, the Zenke Lab LaTeX template, van den Burg's `LaTeXReviewReplyTemplate`, the `bamdone-rebuttal` Typst package, and Editage's fillable form. SIGPLAN's *Guidelines for the Program Chair of a SIGPLAN Event* is the venue-side statement.
- The percentages in section 2 are Table 3 of Kennard et al., *DISAPERE: A Dataset for Discourse Structure in Peer Review Discussions*, NAACL 2022.

Filling the Response Template

Step seven, slot by slot, with the budget written down first

1 Where this sits

The response procedure has nine steps and it is on the website at `resources/research/respond/procedure.md`. Only step seven writes. This file is step seven opened up: which slot of `template_rebuttal.tex` gets filled, in what order, out of which input, and what goes wrong when the step is skipped.

The test for whether a step belongs here is the one the procedure already states. A step whose failure cannot be named is a habit rather than a procedure, and it can be dropped. Every step below names its failure.

2 What the search settled

Question 1 asks whether turning the advice into a template and a fill procedure is encouraged practice. The template half is answered in section 1 of `template_rebuttal.tex`: it is ordinary practice, and every published artifact is a journal response letter rather than anything a 750 word plain text box could hold.

The procedure half answers differently. Nothing published is a fill procedure. The response templates that exist are containers, a macro for a reviewer point and a macro for a reply, running in the order the reviews arrived. None of them supplies an order of work, a word budget, or a test for whether a filled one is finished. The advice runs the other way round, all order and no container. Noble gives ten rules and no skeleton. Sap gives heading practice and no skeleton. The venue calls give a scope sentence and a word count and stop.

So the pair has to be built, and the two halves have to be built against each other. Filling a container nobody wrote a procedure for is where the whole of question 1 sits, and the answer is that it is not discouraged. It is simply not done.

2.1 The three numbers that decide the order

DISAPERE labels every sentence of 506 review and response pairs from ICLR 2019 and 2020. Three of its counts change how this template is filled.

Answering is most of the work. Answering a question is 32.76 percent of all response sentences, three times the next argumentative label. Everything else in a response competes for what is left over.

One request in four asks for a run. Requests are 19.82 percent of review sentences, and inside that, requests for experiments are 4.78 against 14.42 for explanations, clarifications, edits and typos. Most of what a reviewer asks for closes with words.

Two sentences in five of a review are not a comment. Structuring, social text and the uncategorized remainder come to 38.82 percent of review sentences. That is the size of the pile the coverage audit is expected to close on inspection, and it is the reason the audit is cheap enough to run.

3 Six inputs, and what breaks without each

The fill step does not start until all six exist. They are products of steps 1 through 6, not of this one.

Input	What breaks without it
Venue mechanics	The draft is written for a formatted document and rewritten for a plain text box in the last hour, or the local count disagrees with the site's count at the deadline.
The extraction file	The draft argues against a remembered review, and it characterizes prior work instead of quoting it. Both are invisible to whoever wrote the draft and obvious to the reviewer who wrote the comment.
The reviewer cards	The response becomes three single-reviewer responses stapled together, and it gives up ground no comment asked for.
The clusters, each named by its root	Sections collect a topic and commit to none of it.
The classification	A politely worded comment naming a real defect receives a defense.
The gate decisions, signed off	A promise reaches the form that the author never agreed to make, and promises are enforceable.

4 The budget, written down first

The whole allowance is 750 words at ACSAC and at the IEEE Symposium on Security and Privacy. The ACSAC set came to seven clusters. That fixes the arithmetic before a word is drafted.

Seven headings at six words each is 42. One sentence of thanks at the top is another 15. What is left is 693 words over seven clusters, which is 99 each, which is four sentences of 25.

Four sentences. The four roles in the template are not a preference about style. They are what the budget allows: one answer, one ground, one consequence, one change. A fifth sentence anywhere is a sentence taken out of somewhere else.

The allowances are not equal, and they are written beside the headings. Initial score position bounds how far a review can move, so a review near the line is worth more words than one far from it. Deciding that with a number, before drafting, is the only point at which it can be decided on the evidence rather than on how the drafting happened to go.

The scaffolding has a measured size and it gets cut on purpose. At ICLR lengths, structuring, summary, social text, follow-up questions and the uncategorized remainder are 36.54

percent of response sentences. Held at that rate over 750 words, scaffolding would take 274 of them. Here it is seven headings and one sentence of thanks. No summary paragraph, no preview of what each cluster will do, no closing. Fifty-seven words, not 274.

5 The fill order

Eight passes, each across the whole document rather than down one cluster at a time. Free-writing, outlining, sketching, formalizing and drafting are five different activities and should not be attempted at once. The same holds for the six things a cluster has to do.

1. **Number the clusters.** Write the word allowance beside each heading, weighted toward the reviews that can move.

   Skipped: drafting in order spends the budget in order. The first cluster comes out at full depth, the last three come out compressed, and the deepest answer lands wherever drafting started instead of where the movement is.

2. **Title every cluster, then stop.** Fill every `\rcluster` before any body exists. Kind first, subject second. The submitted v2 of this package is the worked set: a misunderstanding about sample efficiency, a clarification of the benefit of opponent modeling, a consideration of the ablation design, the motivation and hidden assumptions of the heuristics, the misleading terminology about simulation and emulation, the editorial changes, the potential experiments.

   Skipped: a title naming only a topic collects everything on that topic and commits the section to none of it, so three questions get answered at a third of the depth each. That is inadequate explanation, the largest negative in the validated feature set at 0.31.

3. **Paste every echo.** Fill every `\recho` with the reviewer's own characters, copied out of the extraction file, never retyped from memory. A phrase longer than about ten words is not the distinctive part of it.

   Skipped: a correct sentence the reader cannot attach to their own question is a true sentence answering nothing, and in a document read by three people with different concerns, nobody does the work of matching it.

4. **Run the gate before any change slot holds text.** Every request takes the three questions in order and comes back signed off, and only then do `\rchange`, `\rcommit` and `\roffer` get filled. Three requests in four ask for an explanation, a clarification, an edit or a typo fix, and those close inside the response.

   Skipped: a commitment nobody agreed to make is discovered at export, with no time left to withdraw it. A program committee can reject a camera ready that fails to deliver what the response promised.

5. **Answer every cluster before grounding any of them.** Fill every `\ranswer` slot in one pass, direct and as asked, graded to the size of the error: yes, no, not quite, almost.

   Skipped: grounding a cluster whose answer is still unwritten produces a paragraph of background with the answer at the bottom, or with no answer in it. A reviewer scanning for whether their point was addressed should not have to read to the end to find out.

6. **Ground, then consequence.** `\rground` carries the fact, the number, the equation or the passage. `\rconseq` carries what follows, formally and for whoever would use the system. A number goes in only where it changes a decision.

   Skipped: an explanation that stops at the mechanism leaves the reviewer to work out the implication, and the implication is the part they asked about. A number with no denominator and no use is read as evidence and displaces evidence that would have done work.

7. **Audit against the reviews, not against the draft.** List every reviewer sentence that no cluster quotes, then sort the list into summary, praise and gap. Most of it closes on inspection. What survives gets a clause inside an existing cluster, or the one coverage sentence saying the concerns were read, are agreed with, and will be addressed in a revised version.

   Skipped: auditing the draft against itself confirms only that the draft says what it says. Silence on a reviewer's question, however minor, is what the venues and the evidence both read as evasion.

8. **Export, count, hold.** Generate the plain text from this source rather than keeping a second copy by hand. Check that no macro, marker, identifier, link or gate survived. Count on the venue's own counter and treat it as authoritative. Over the limit, cut restatement, then spectator voice, then pointing, then the cluster answering the least movable review. Then it goes to the author, and the author submits.

   Skipped: a second copy kept by hand drifts from the first, and the drift is found at the form, in the last hour, with no way to tell which version is the real one.

6 When the venue says nothing

Two silences, and they take opposite branches.

Silence about shape. No venue says how a response should be organized. The strongest statement from a venue body is SIGPLAN's guidance to its chairs, which encourages holding a response phase and stops there. That silence is a free choice, and the choice goes to the reader who is skimming. What reaches the decision is a chair's synthesis of whether the comments were met, so structure a chair can skim beats prose that repays a close read.

Silence about content. ACSAC does not say whether a new number may appear. The IEEE Symposium on Security and Privacy does say, and forbids new results outright in the non-interactive rebuttal. Where the venue is silent and its neighbor is explicit, the explicit rule is the safe reading and following it costs almost nothing. Security custom fills the ACSAC silence the same way: a small new number answering a reviewer's question is normal, a new topic is not.

Write down which branch was taken and why, in the package, so the next submission to the same venue does not argue it a second time.

7 Exit tests

Ten checks over the filled source. All of them are mechanical, which is the point. A check run as an intention is not run.

- Every `\rcluster` names a kind before its subject.

- Every cluster has exactly one `\recho`, and it sits in the first clause rather than in a sentence of its own.
- Every cluster has an `\ranswer`, and nothing precedes it but the echo.
- Every `\rcommit` names a configuration and the table it lands in.
- Every `\roffer` is numbered and says what it would show.
- No `\rchange` carries work that did not come back through the gate.
- No sentence is left with no reviewer comment behind it.
- Every reviewer question is either quoted inside a cluster or on the audit list with a decision written beside it.
- The venue's own counter reads under the limit.
- The exported plain text contains none of the macros, and none of the identifiers.

8 Sources

- The nine steps, the gate, the classification, the clustering rule, the four sentence roles, the phrasing shelf and the cut order are the author's own, at `content/resources/research/respond/` on the website.
- The cluster titles in step 2 are the section headings of `v2.tex`, the submitted response in this package.
- The sentence counts are Tables 2 and 3 of Kennard et al., *DISAPERE: A Dataset for Discourse Structure in Peer Review Discussions*, NAACL 2022.
- The odds ratio 0.31 for inadequate explanation is from *Rebuttals Move Peer-Review Scores, but Initial-Review Structure Bounds the Movement*, arXiv 2606.22166, 2026.
- The 750 word limit and the ban on new results are from `sp2027.ieee-security.org/cfpapers.html` and the ACSAC calls for papers. The venue-side statement on response phases is SIGPLAN's *Guidelines for the Program Chair of a SIGPLAN Event*.

Comment Catalog Template

One row per comment, and the options still open

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1 What the catalog is for

Question 68 gives the shape. After the comments have been labeled and cataloged, a table whose first cell is the comment, carrying a link that jumps to one or more complete excerpts, then several columns of characterization, then one cell chosen from a list rather than written, then a few ways of answering that comment or that group of comments.

The catalog sits between reading the reviews and drafting the response, and it does the one thing a draft cannot. It holds every comment open at the same time. A response written straight from the reviews settles the first comment before the last one has been read, and the answer that reads well inside a paragraph is not always the answer that survives the whole set. So the catalog fixes the order of the work: characterize everything, list the ways of answering, choose last.

It is working apparatus. None of it reaches the response form.

2 The columns

Eight, in this order. Three of them carry the characterization, one is the selected cell, and the last two are the part the author decides.

Column	What it holds	Values
Comment	The identifier, then the comment in one line. Use the reviewer's own words where one sentence carries the whole complaint, and a compression that keeps their nouns where it does not. The identifier is the link.	The identifier, then free text
Excerpts	Where the complete text lives. The response cluster that answers it, and every manuscript passage the comment is about, by file and line.	Free text, locators only
Disputes	What the comment challenges.	Fixed list
Stance	What we do with it.	Fixed list
Root	The other identifiers that would be answered by the same change. Empty when the comment travels alone.	Identifiers
Cost	What answering it would take. This is the selected cell.	Fixed list
Options	Two or more ways of answering, most preferred first, one line each.	Numbered free text
Decision	The option chosen, or the word open and the name of what settles it.	Free text

Why Root is a column and not a grouping. Grouping the rows by cluster hides the comments that belong to two clusters at once, and those are the expensive ones. Keep one row per comment, in reviewer order, and let the Root cell carry the grouping. The response clusters are then read off the column instead of being imposed on the table.

3 The three fixed lists

Disputes. What the comment challenges, in the vocabulary of claims and attacks. A challenge to the CLAIM disputes what the paper asserts. A challenge to the EVIDENCE accepts the claim as meaningful and disputes what supports it. A challenge to the WARRANT accepts both and disputes the step between them. A challenge to a DEFINITION disputes what a term picks out. A challenge to SIGNIFICANCE disputes whether the result matters. PRESENTATION is the sixth value and covers everything that disputes neither the argument nor the result. A strength gets the value NONE here and in the two columns after it, and it gets a row, because a strength is a thing the response can lose.

Stance. REFUTE, the comment rests on a misreading. CONCEDE, the comment finds a real defect. NARROW, the evidence supports something smaller than the paper claims and the claim moves. REFRAME, the fact is not in dispute and its presentation is. DEFER, real and outside this window. Two values in one cell are allowed and are common: a comment that bundles a factual reading with an inference drawn from it is refuted on the reading and reframed on the inference, and writing both into the cell is what keeps the two halves from being answered as one.

Cost. WORDS ONLY, answerable from what is already written. EXISTING DATA, answerable from runs already performed, including reporting them differently. ONE RUN, one configuration under the protocol already used. FOLLOW-UP PROJECT, work at a scale that a revision window does not hold.

4 The one selected cell

Cost is the cell chosen from a list rather than written, and it sits between the characterization and the options because it decides which options are admissible before any of them is drafted.

The rule it enforces is short. A row whose cost is WORDS ONLY or EXISTING DATA takes a commitment. A row whose cost is ONE RUN takes a commitment only with the author's signature on it. A row whose cost is FOLLOW-UP PROJECT never takes a commitment, because promising work at that scale says the paper is a major revision at best, and the venue's own categories then apply that reading rather than ours.

Filling this cell honestly is the whole value of the column. The temptation is to write WORDS ONLY for a comment that a paragraph could deflect, and the check against it is the excerpt column: if no passage in the manuscript already contains the answer, the cost is not words.

5 The options column

Two or more, most preferred first, each one line, each naming what it does and what it costs. The last option is always the one that answers without new work, so the row can be closed under time pressure without being reopened.

Several options are required when the comment is a question asking for work. Question 68 gives the reason: comments arrive in the form "could the authors evaluate attackers that differ in other behaviors," and a single option turns that question into a commitment before the author has seen the choice. Three options are usually available for such a comment, and they are worth writing out even when the first is obviously right. Answer the question as asked, from what exists. Offer the work, numbered, with what it would show. Commit to the work outright.

An option is not a preference until it says what it buys and what it gives up. One line for each is enough, and the line goes in the option, not in the decision. The decision cell records the choice. It does not re-argue it.

The option nobody writes down is the one worth adding. A comment answered by a correction rather than by a run appears only if someone checks whether the run in question was already performed under a different label. Put that check in the excerpt column and the option writes itself.

6 The link

The first cell links to the full excerpt so that the table can be read without the reviews open beside it. In a compiled document the mechanism is a target and a link, one target per comment, placed in the excerpt box.

The target goes once, where the reviewer's complete sentences are quoted. The identifier in the catalog carries the link. Both stay stable across drafts, which matters because the identifiers themselves are an editorial invention and appear nowhere in the reviews.

7 The macros

```
% One catalog row. Eight cells, in the column order above.
\newcommand{\ccrow}[8]{%
  \ccid{#1} & #2 & #3 & #4 & #5 & \cccost{#6} & #7 & #8 \\}

% The link. \cctarget{A4} goes once, in the excerpt box that quotes A4
% in full. \ccid{A4} renders the linked identifier in the catalog.
\newcommand{\cctarget}[1]{\hypertarget{cc#1}{}}
\newcommand{\ccid}[1]{\hyperlink{cc#1}{\textbf{#1}}}

% The selected cell. One of: words only, existing data, one run,
% follow-up project. Anything outside the list is a value the list
% does not have, and the list is what makes the column sortable.
\newcommand{\cccost}[1]{{\scshape #1}}

% One option. #1 its number, #2 what it does and what it costs.
\newcommand{\ccopt}[2]{\textbf{#1.}\ #2}

% A decision that has not been made. #1 what settles it.
\newcommand{\ccopen}[1]{{\itshape Open.}\ #1}
```

8 The skeleton, one row

```
\ccrow{[identifier]}
{[the comment in one line, their nouns kept]}
{[response cluster; manuscript file and line, one per passage]}
{[claim / evidence / warrant / definition / significance / presentation]}
{[refute / concede / narrow / reframe / defer]}
{[identifiers sharing the root, or empty]}
{[words only / existing data / one run / follow-up project]}
{\ccopt{1}{[what it does, what it costs]} \ccopt{2}{[the fallback that needs no new
work]}}
{[the option chosen, or \ccopen{[what settles it]}]}
```

The table itself is a `longtable` on a sideways page, set by widening the geometry around the table alone and restoring it afterward, so the prose sections keep a readable measure. Eight columns do not fit upright at a size anyone would read, and shrinking the options column until they do defeats the column.

9 Filling it

Six steps, and only the last two are decisions.

1. Extract every comment from the reviews verbatim, including the strengths. Assign each an identifier, reviewer letter and running number, and never renumber.
2. Quote each one in full in an excerpt box and put the target there. One box per comment, in the reviewers' own order.
3. Find every manuscript passage the comment is about and record file and line. This step, not the reading of the review, is where a misreading gets caught.
4. Fill the three characterization columns for all comments before writing a single option. A stance chosen while the next comment is still unread is a stance chosen twice.
5. Write the options. Most preferred first, the no-new-work answer last.
6. Take the decisions, or write what each one waits on. Rows waiting on a verification name the file or the log that would settle them. Rows waiting on the author name nothing else.

10 The spreadsheet version

Question 69 asks for it, and the reason it exists is sharing: a coauthor sorts by cost and reads only the rows that need a signature.

One header row, one row per comment, the same eight columns and the same order. The three fixed lists become validation lists on their columns, which is what makes the selected cell a selection rather than a phrase somebody typed twice two different ways.

Two cells travel badly and both have the same fix. The options cell holds the option labels only, numbered, one per line inside the cell, and the full text of each option stays in the document. The excerpt cell holds the locators without their links, since a link into a compiled document does not survive the export. Everything else transfers one cell to one cell.

Generate the delimited rows from the same source as the table rather than keeping a second copy by hand. A second copy drifts, and the drift is found by whoever trusted the shared file.

11 What the catalog is not

It is not the response, and it is not a stage of the response. The response carries no identifiers, no stance labels, no cost classes, and no options, since options shown to a reviewer read as an invitation to pick the expensive one.

It is also not a record of what was decided after the fact. A catalog written once the response is drafted is a summary of the draft, and it can only agree with it. The value is in the interval where the options are still open, and that interval closes early.

12 Sources

- The column list, the link requirement, the selected cell, and the rule that several options are needed for a comment asking for work are all question 68, in the author's own statement of it.
- The spreadsheet version and its purpose are question 69.
- The claim, evidence, warrant, definition, and significance vocabulary is The Craft of Research, applied to reviewer comments in this package's analysis file and kept in the same senses here.
- The stance list and the root-cause grouping are the author's, already in use in the same file and in the response procedure on the website.
- The cost list is cut at the venue's own boundaries: what a revision window holds, and what it does not.

Experiment Regime Template

Twelve questions answered before the first run, and the answers for this paper

1 What decides the design

Question 16 asks whether a 2×2-like design is necessary and what goes into that decision. One belief decides it, and in most papers that belief is already written down in the method section.

A ladder of ablations identifies the contribution of each component only when the components are additive and independent. A full factorial is the only design that estimates the interaction between them. So the question is never whether the full factorial is more thorough. It is whether you believe the two components affect each other, and you have usually said so already.

The trap is that the sentence which sounds like a reason to skip the fourth cell is the reason to run it. *The components are nested. Ablating B while keeping A does not make sense. Without B the factoring would not be clean.* Each of those says the effect of A is different depending on whether B is present, and that is what an interaction is. The belief that licenses the cheaper design is the opposite one, and it is the stronger claim: that the two components do not touch each other at all.

2 The questionnaire

Twelve questions in four blocks, answered in order, before any configuration is launched. "Not applicable" is an acceptable answer. "We will see" means the design is not chosen yet.

The claim

1. What is the claim, in a sentence a number could contradict? A claim no number can contradict does not need an experiment. It needs a rewrite.
2. Is the claim about why something works, or about which method ranks first? Why-claims survive a change of environment; a ranking on one testbed rarely does, so a design built to produce a ranking returns less than it costs.
3. Which comparison would a reader accept as contradicting the claim? Name the comparison, not the table it would sit in.

The factors

4. List every factor that could change the number, and mark each one varied, held fixed, or uncontrolled. Uncontrolled is a real category, and leaving it off the list is how a result stops replicating.
5. For each varied factor, its levels, and why those levels and not others.
6. For each fixed factor, the level it is pinned at, and the sentence the paper therefore cannot write. A held-fixed factor is a scope limit, and it reads better stated than discovered.

7. Which pairs of factors do you believe interact? Write each belief as a full sentence in the form *the effect of A is different when B is present.*

The design

8. Does any pair from question 7 contain two factors you are claiming credit for? Those pairs need every combination.
9. Does any pair contain one factor you claim credit for and one factor you are quantifying over? Then either the design crosses them, or the claim is conditional and says so.
10. What does one cell cost, in wall-clock and in whichever resource is actually scarce? For a live testbed that resource is rarely compute.

The runs and the report

11. How many seeds per cell, which statistic goes in the table, and what is released for a reader who wants a different statistic?
12. What value do you predict for each new cell, written before the run? A cell reported after the fact is a data point. A cell whose value was predicted from the mechanism is a test of the mechanism.

3 The decision rule

No pair in question 8. The ladder is sufficient, and the additivity belief goes into the paper as a stated assumption, because that is what the ladder assumes.

One pair in question 8. The full factorial over that pair. Two binary components give four cells, and three of them are usually already run.

Three or more claimed components. A resolution-IV fractional factorial recovers the main effects unconfounded with two-factor interactions at roughly half the runs.

Factors quantified over are crossed by the scope of the claim, not by this rule. If the claim says "against both attacker profiles," both appear in every reported cell.

One more line, because it is the one that gets skipped. A held-fixed factor whose level was chosen for convenience is a limitation. A held-fixed factor whose level was chosen because varying it is the follow-up project is a contribution boundary. Say which.

4 Filled, for this paper

Factor	Status	Levels	Why
Exogenous factoring	varied, claimed	on, off	first technical contribution
Opponent model	varied, claimed	on, off	second technical contribution
Learning method	varied, contrast	FOE-Dreamer, Rainbow, IQN, Expert, Random, Sleep	the sample-efficiency claim and the operational floor
Attacker profile	quantified over	Avoidant, Enticed	the claim is made against both, so both appear in every row
User process	varied in the diagnostics only	U-rand, U-burst	the two linear-decoding tables
Seed	5 per cell	5	seed-to-seed variance
Topology	fixed	8 hosts, 3 subnets	offered as a follow-up run
Defender profile	fixed	one	offered as a follow-up run
Episode structure	fixed	100 steps, 5 s polling	fixes what one interaction costs
Training budget	fixed	3 days, 1 GPU, 8 parallel environment instances	the live budget the claim is made under

Question 7, the interaction belief, is already written and already submitted. The response says the full 2×2 "was considered and set aside because the components are nested; without opponent modeling the factoring would not separate user-driven from attacker-driven variation, so the Factor+No-Opponent cell would not evaluate the factoring as designed."

That sentence answers question 8 in the affirmative. It says the factoring behaves differently depending on whether the opponent model is present, which puts the pair into the design rather than out of it. The identifiability argument in `f01` says the same thing in stronger terms: the privileged attacker supervision is the third auxiliary signal that makes the exogenous latent identifiable as user activity rather than as uncontrollable reward-relevant variation, so removing the opponent model removes the condition under which the factoring's interpretation holds. The reviewer asked for the one cell that becomes informative exactly when the nesting belief is true.

Question 12, the prediction, before the run. Read against the Avoidant column of the ablation table, additivity predicts $-78 + 16 = -62$ for Factor with no opponent. The nesting belief predicts worse than that, somewhere between -62 and -78. The sharper test is not the loss at all: fit the same ridge regression from u_t to attacker progress rather than to user activity, in both configurations, and the nesting belief predicts that it rises when the opponent model is removed. That measurement needs no new training run for the configurations already trained.

5 What the design commits you to reporting

Choosing the design settles which rows exist. It does not settle what goes in them, and the second choice is where papers in this area lose credit.

Seeds. Five per configuration is what this project has, and five is a floor rather than a target. The security systematization of deep reinforcement learning work sets a minimum of five and encourages twenty or more, and the RL methodology literature reports a case where a five-seed study of a known algorithm gave a higher mean and tighter error bars than the same algorithm at thirty seeds. Five seeds with the per-seed numbers released is defensible. Five seeds with only a mean and a standard deviation is not.

Statistic. Mean and standard deviation over seeds is what the tables carry. What makes that usable is the artifact bundle, which carries the raw per-seed episode losses so a reader can compute a bootstrap interval or an interquartile mean without asking for anything.

Convergence. The learning curves across the three-day budget are the evidence that training was not stopped early, and they belong in the body rather than an appendix. Forty-two percent of the sixty-six papers in the security systematization show none at all.

Hyperparameters. The same list every time: algorithm settings, network shape, optimizer settings, and the training procedure including the number of parallel environment instances. Thirty-two percent of that corpus reports none.

What the design is called. Five seeds on a live testbed is not a benchmark ranking and should not be written as one. It is a deployment study, which the RL methodology literature names as the setting that receives the least methodological guidance, and the setting where a paired difference against a matched baseline is worth more than a leaderboard row.

6 The cheapest sufficient version

When the design the rule selects does not fit the window, three reductions in this order, and only the third costs anything.

Measure before you train. The diagnostic that separates the two beliefs here is a regression on stored latents, not a new configuration. Any question answerable from trajectories already collected gets asked first.

Cut levels of a factor you are quantifying over before you cut a cell of a factor you claim credit for. A claim against one attacker profile is smaller and still true. A missing cell leaves the interaction unestimated, and the paper's own architecture says it is there.

Cut seeds last, and when you do, release the per-seed numbers and call the result what it is.

7 Sources

- The factorial argument, the one-factor-at-a-time defense and its four preconditions, and the resolution-IV fallback are written out in `ablation_design.tex` in this package, with its own sources: the standard design-of-experiments treatments, Frey and Wang on when one factor at a time can win, and Collins et al., *Design of Experiments with Multiple Independent Variables*.
- The identifiability argument behind the interaction belief, and the four assumptions with the two that fail, are in `f01` in this package. The predicted values for the missing cell are in `f05`.
- Patterson, Neumann, M. White and A. White, *Empirical Design in Reinforcement Learning*, JMLR 2024, for the seed-count evidence, the deployment setting, the calibration baselines, and the instruction to fix the hypothesis before the runs.

- McFadden et al., *SoK: The Pitfalls of Deep Reinforcement Learning for Cybersecurity*, for the prevalence figures and the reporting floors over its sixty-six-paper corpus.
- Agarwal, Schwarzer, Castro, Courville and Bellemare, *Deep Reinforcement Learning at the Edge of the Statistical Precipice*, NeurIPS 2021, for the aggregate statistics that a handful of seeds can still support.

Baseline Choice Template

One baseline per claim, seven roles, and a gate before adding an eighth

1 What a baseline is for

A baseline answers one question: what would you have got without the thing you are claiming credit for? Different baselines answer different versions of that question, and a paper needs the ones matching its claims. That is the whole principle, and everything below is its consequence.

Two failures follow from ignoring it, and they point in opposite directions. A paper with a baseline no claim needs has spent runs on a comparison nobody asked for. A paper with a claim no baseline tests has an assertion sitting where evidence should be, and a reviewer will find it before a reader does.

2 Seven roles

The taxonomy question 21 asks for is not a taxonomy of algorithms. Rainbow and a vanilla Dreamer agent are different kinds of algorithm but they can hold the same role, and one algorithm can hold two. The useful division is by the question the comparison answers.

Role	The question it answers	This paper
Floor	What does no skill at all get in this environment?	Sleep, Random
Operational reference	What does current practice get, without learning anything?	Expert-Inspired Heuristic
Paradigm contrast	Is the gain the family of method, or this method?	Rainbow, IQN
Recipe ancestor	Is the gain the family, or my additions to it?	the no-factoring, no-opponent cell, which is a vanilla Dreamer agent
Component ablation	Which addition, and do the additions interact?	the 2×2, one cell short
Confound control	Is the gain the extra information rather than the architecture?	absent
Ceiling	What would the same information buy if it were simply handed over?	absent

Three of these are the ones security reviewers ask for by name and machine learning venues often skip. The floor is how a reader knows what the numbers mean at all. The operational reference is what a practitioner compares against, and a learned defender that does not beat the deployed playbook has no case for being deployed. The confound control is the one that decides whether a result is believed.

The recipe ancestor is the role most often filled without being labeled. If the bottom row of an ablation table is the standard agent, saying so converts an ablation into a paradigm comparison at no cost, and a reviewer who asks for a vanilla baseline is asking for that sentence rather than for a run.

3 The map, one row per claim

```
Claim, as a sentence a number could contradict
Role that tests it
The arm that fills the role
What is held equal between the arms
What is not equal, and is stated instead
Present, missing, or filled by an existing row under another name
```

Write the claims first, from the abstract and the contribution list, and only then the arms. Doing it the other way produces a table of everything that was run, which is a different document.

The last field is the one that saves runs. A missing row is often an existing row that has not been named, and naming it is a writing change.

4 Principles

One baseline per claim. A comparison no claim needs is not evidence, it is a run you have to defend.

Match the resource the claim is about. A sample-efficiency claim is quantified in environment interactions and has to report them. A deployment-cost claim is quantified in wall-clock on the target system. The two order methods differently, and holding one fixed does not hold the other fixed.

Match the mechanism, not the information. A baseline that receives the same bits through a different pathway tests something else. Predicting a signal forces the representation to encode structure related to it; observing the same signal as an input feature does not. The two uses are not interchangeable, and a reviewer can say the variant answers a question nobody asked.

Tune the baseline as hard as your own method, and say how. Default settings carried over from another environment are the standard way a comparison goes wrong, and the fix is cheaper than the run it discredits.

Prefer the baseline the field already recognizes. A bespoke variant that loses proves less than a standard one that loses.

In security, the floor and the operational reference are not optional. The systematization of deep reinforcement learning work in this area finds that half the corpus does not separate the contribution of the learned policy from the contribution of how the task was set up, and the cure it names is an evaluation under random action sampling.

Add a baseline only when it removes a confound nothing else removes. Not when it merely adds a comparison.

Weigh the asymmetry before promising a run. A confirming result restates what a corrected sentence already says. A disconfirming result damages the headline claim, and during a shepherded revision there is no room left to reconsider.

5 Model-based against model-free

The like-for-like question has a clean answer and an inconvenient one.

The clean answer is the list of things that must be identical: the observation stream, the action space, the reward, the episode structure, the seeds, and the effort spent searching for hyperparameters. Anything on that list left unequal is a confound, and the fix is a run rather than a sentence.

The inconvenient answer is that two items can never be made equal, so they get stated instead of matched. The first is the number of gradient updates per environment step, which is what the model-based method is buying with its extra compute; equalizing it would remove the mechanism under study. The second is the auxiliary supervision each method receives. The rule for the second comes from the security systematization and is worth using verbatim: comparisons between approaches should be performed with equivalent information and capabilities.

Which leaves the axis. The resource a model-free agent spends is environment interactions. The resource a model-based agent spends is compute per interaction. A fixed wall-clock budget on a live testbed is a deployment-cost comparison, and it is the right comparison for a paper about training in the operational network. It is not by itself a sample-efficiency comparison, and a paper claiming sample efficiency has to report the interaction count for every arm. That is one number per arm, recoverable from logs, and it answers the reviewer who asked what the episode count actually counts at the same time.

6 The gate for adding one

Three questions, in order, before any new arm enters a response or a revision.

1. Which role in the table above is empty, and which claim is left untested by its emptiness? If no claim is untested, the answer is a sentence naming the existing row, not a run.
2. Would the new arm test the mechanism, or only the information? If only the information, name the mechanism-matched version and offer that one instead.
3. Is there a cheaper measurement that answers the same question from the other direction? A surviving log from an earlier attempt answers a confounding worry at the cost of a search rather than an experiment.

What leaves the gate is one of three things: a commitment naming a configuration and the table it lands in, a decline stating a reason that is never effort and never the word limit, or a numbered offer with what it would show.

7 Filled, for this paper

Claim	**Arm**	**State**
Model-based learning beats model-free under a fixed live budget	Rainbow, IQN	present; the budget matched is wall-clock, so the interaction counts are the missing number
The learned defender exceeds current practice	Expert, Random, Sleep	present
The two additions beat the plain recipe	the no-factoring, no-opponent cell	present, and equivalent to a vanilla Dreamer agent; saying so is the whole answer to the comment
Each addition contributes separately	the 2×2	one cell short, and the missing cell is the informative one
The gain is the architecture, not the privileged supervision	none	untested

The last row is the one the comment about baseline fairness points at, and it is worth being precise about what the comment says. It names a writing defect: one section states that the model trains with ground-truth attacker labels, another asserts that no baseline receives privileged information the model does not also see, and the second sentence runs the wrong way to establish equivalent supervision. Replacing that sentence with an account of what each method received during training, and what remains available at deployment, fixes what broke.

Three candidates could fill the row, in increasing order of what they cost and decreasing order of how likely they are to be needed. A search through the early model-free attempts, which answers the confounding worry from the other direction if the logs survived. Rainbow with the same auxiliary prediction head, which matches the mechanism. Rainbow with attacker state appended to its observation vector, which matches the information and not the mechanism, and is therefore the one to decline.

8 Sources

- The claim-to-baseline table for this paper, the reading of what the comment about baseline fairness actually asks for, and the verdict on the appended-feature variant are in `baseline_choice.tex` in this package.
- McFadden et al., *SoK: The Pitfalls of Deep Reinforcement Learning for Cybersecurity*, for the equivalent-information rule, the random-action ablation, and the requirement that applying this family of method be justified against the in-domain alternative rather than only against other agents.
- Patterson, Neumann, M. White and A. White, *Empirical Design in Reinforcement Learning*, JMLR 2024, on calibration baselines, designer bias, and tuning the baseline as hard as your own method.
- Henderson et al., *Deep Reinforcement Learning that Matters*, for the worked case where retuning a baseline reverses the result.

- Arp et al., *Dos and Don'ts of Machine Learning in Computer Security*, USENIX Security 2022, for the inappropriate-baseline pitfall as the security community states it.
- The reviewer comments quoted here are reproduced in the reviews file of this package.

Agent Profile Template

One sheet per agent, and the three checks a pair of profiles has to pass

1 Whose words to use

Question 44 and question 38 have one answer between them. The security community already has a settled vocabulary for both halves, attacker and defender, and it is not the vocabulary the reinforcement learning side uses. A reviewer who works in security reads our words as looseness, and they are right to. This table is the mapping.

What we say	What they say	Where it is fixed
Opponent type	Threat actor, characterized by sophistication and resource level as two separate properties	STIX 2.1, Threat Actor object
Scripted opponent	*Adversary emulation* when it reproduces a documented actor, *red teaming* when it searches freely for a way through	MITRE ATT&CK, and this project's own threat modeling section
Opponent action set	Tactics, techniques, and procedures	ATT&CK
Defender priorities	Security posture, set by a risk appetite and made operational by a risk tolerance	CNSSI 4009-2022, and NIST CSF 2.0 at GV.RM-02
Defender action set	Defensive tactics: Model, Harden, Detect, Isolate, Deceive, Evict, Restore	MITRE D3FEND
Deception actions	Adversary engagement, with goals Expose, Affect, and Elicit	MITRE Engage

One term to stop using loosely. Question 38 says attacker emulation. The term of art is *adversary emulation*, and it means something narrower than we mean. An emulation plan starts from cyber threat intelligence on a named actor and reproduces not only what that actor does but which tools and commands it does it with and at which stage of a breach. Emulation is a claim about provenance, not a claim about fidelity. A profile invented for a testbed can be sharper, more separable and better controlled than any documented actor, and it is still not emulation.

2 The attacker sheet

Nine fields. Every one of them is short. A field left blank is a field a reviewer fills in for you.

Name and class. The label used in every table and figure, fixed once.
Sophistication. An ordered scale, running from a complete novice to an actor able to influence supply chains in order to introduce vulnerabilities. STIX runs it as none, minimal, intermediate, advanced, expert, innovator, strategic.
Resource level. A second axis, deliberately separate from the first: individual, club, contest, team, organization, government. They are separate because a highly skilled actor may have almost nothing behind it while a minimal one may have the resources of an organized crime ring.
Goals. What this actor wants out of this network, named in the network's own objects rather than in reward.
Primary motivation. One value. Secondary motivations go on a second line or nowhere.
Techniques. The action set, mapped to ATT&CK wherever a mapping exists, and marked as unmapped where none does.
Decision rule. What the actor does with what it sees. This is the field that makes it an agent instead of a trace, and it is the field the opponent model is trying to recover.
Observability. What the actor can see. Probes belong here, with what each one returns.
Provenance. A documented actor and the intelligence that documents it, or invented for this testbed and the behavior it abstracts. One or the other. There is no third answer, and writing neither is how a paper ends up claiming emulation it did not do.

3 The defender sheet

Question 44 wants defenders with different priorities and different postures. Posture is the right word and it has a definition: "the security status of an enterprise's networks, information, and systems based on CS resources (e.g., people, hardware, software, policies) and capabilities in place to manage the defense of the enterprise and to react as the situation changes." Six fields.

Posture, in one sentence. What this defender is protecting and what it is willing to lose to protect it.
Risk appetite, then risk tolerance. Appetite is how much risk the organization will accept in pursuit of an objective. Tolerance is the operational limit inside that appetite, and in an experiment it is a number in the reward function. Write the number first and the reward second. Doing it the other way around means the posture is whatever the reward happened to imply.
Permitted tactics. Which of the seven defensive tactics this profile may use at all. Two agents that differ only in the weights on one reward are the same profile trained twice. Two agents that differ in which tactics they may reach for are two profiles.
Engagement goal. Evict, or elicit. On one host, at one time, it cannot be both, and choosing between them is the sharpest difference there is between two real defenders. It is also the choice the deception action makes.
Observability. What it sees and at what rate. Verbosity is an action in most action sets, so state the default and state what raising it costs.
What it will not do. The action this profile declines, and the reason. A profile with nothing in this field is an optimizer, not a posture.

The action space, said in their words

Our action	D3FEND tactic	What it commits the defender to
Isolate	Isolate	A barrier, at the cost of the traffic behind it
Disrupt	Evict, Restore	Removal of in-memory persistence, and of live sessions
Monitor	Detect	Fidelity later, paid for now
Deceive	Deceive	Engagement, which is Expose if the alert is the point and Elicit if the attacker's next technique is
Null	None	The decision to spend nothing this step

The two tactics with no action in this space are Model and Harden, and both of them happen before an episode starts. Saying so in the threat model costs a sentence. Being asked for it costs a review cycle.

4 The three checks

These are what question 38 asks for when it says double check our justification. Run them on the pair, not on either profile alone.

1. Name the dimension before you name the profiles. Two profiles are a pair only if you can say in one clause what varies between them. If the clause is hard to write, the pair was chosen by what was easy to script.

2. The separating dimension must not be the one the method reads. This is what Reviewer C asked, in C8: "The attacker profiles may favor the proposed opponent model. Avoidant and Enticed mainly differ in how they respond to deception signals, which directly matches the defender's deception actions. Could the authors evaluate attackers that differ in other behaviors to show that the benefit of opponent modeling is not specific to this setup?" When the dimension that separates the profiles is the dimension the encoder is built to recover, the experiment measures the pair rather than the method. The fix is a second pair varying along something the encoder does not see, reported in the same table as the first.

3. A bracketing claim needs the study that fixes the bracket. Saying two profiles bracket a spectrum observed in field studies is a claim about the field studies, and it is checkable. Either cite them and name both endpoints in their terms, or drop the word and say what is true instead: the profiles span the decision rule applied to deception outcomes, and nothing else.

5 The macros

Paste these into the preamble of whatever the profiles are being written into. They mark the fields the coverage check looks for, and they typeset as plain text.

```
% One sheet. #1 attacker or defender, #2 the profile name.
\newcommand{\apsheet}[2]{\subsection*{#1 profile: #2}}
```

```
% The fields. Kept as macros so a missing one is greppable.
\newcommand{\apfield}[2]{\textbf{#1.} #2}

% The separating dimension of a pair, stated once, before either profile.
\newcommand{\apdim}[1]{#1}

% Provenance. Documented actor plus source, or invented plus what it abstracts.
\newcommand{\apsrc}[1]{#1}

% The action the profile declines, and why. Blank fails the defender sheet.
\newcommand{\apdecline}[2]{#1, because #2}

% Risk tolerance as a number, written before the reward that encodes it.
\newcommand{\aptol}[2]{#1 per #2}
```

6 Filled, on this project

The attacker pair. Avoidant and Enticed sit at the same point on any sophistication scale, because both run the same three deterministic binary probes over the same attacker-observable information. Resource level is not stated for either and does not need to be, since neither profile spends anything. The separating dimension is the decision rule applied to the probe outcomes, conservative against permissive. Provenance is invented for this testbed, abstracting the honeypot fingerprinting work the banner probe is built on, and it is not emulation of a named actor. Check 2 is the one that costs us: the separating dimension is exactly the signal the opponent encoder reads, which is why C8 landed.

The defender pair that would answer it. An evicting defender isolates and disrupts on the first alert that clears threshold, never deceives, and accepts dropped sessions as the price. An eliciting defender leaves the honeypot standing while the attacker is inside it, pays the alert cost, and declines to isolate until the attacker has revealed a second technique. They differ in engagement goal, in permitted tactic, and in tolerance for user-visible disruption, and that third one has to be a number in the reward before either is trained. Neither of them differs from the other in a way the opponent encoder was designed to read, which is the point.

7 Sources

- The threat actor property set, the sophistication scale and its endpoints, the resource level scale, and the argument for keeping the two apart are from the STIX 2.1 specification, Threat Actor object and its vocabularies.
- The adversary emulation definition, the emulation plan structure (intelligence summary, operations flow, step by step procedures in human and machine readable form, infrastructure), and the split between full and micro plans are from MITRE ATT&CK and the Center for Threat-Informed Defense adversary emulation library.
- The seven defensive tactics are MITRE D3FEND. The engagement goals Expose, Affect, and

Elicit, and the surrounding Prepare and Understand, are MITRE Engage.

- The security posture definition quoted in section 3 is CNSSI 4009-2022, as carried in the NIST glossary. The risk appetite and risk tolerance distinction is NIST CSF 2.0 GV.RM-02 and NIST IR 8286A.
- The separation between adversary emulation and red teaming used here is the stipulative one already written into this project's own threat modeling section, where emulation is the narrower case that models a particular adversary's known behavior.
- NIST SP 800-30 Rev. 1 assesses an adversarial threat source along capability, intent, and targeting. Read through a search summary rather than from the publication, so it is marked secondary and should be confirmed against Appendix D before being cited.
- C8 is quoted from the review email of 19 August 2026. The probe description, the profile decision rules, and the five element defender action space are from the threat model and testbed section of the paper.

Project Design Template

One sheet, three gates, and four tests that say whether to start

1 What is actually missing

Question 34 wants a detour out of one project to make a connection to neighbouring methods concrete, and its note says the process of project designing should be standardized first. The research statement already names the shortage precisely: it is not a shortage of projects, it is a shortage of sharpened arguments connecting challenge, method, and contribution. Ideas arrive in free-form brainstorming and stay there. The sheet below is the shape they get poured into.

Two design decisions before the sheet, both taken from the Think page.

A list of heuristics is not the deliverable. General heuristics transfer badly, because knowing an operation and regulating when to use it are different capacities. So this is a procedure with stopping conditions rather than a list of moves, and the stopping conditions are the part that does the work.

Read the stage list as a menu, not a mandate. The useful move is to walk it once at the start of a project and write down which stages you are consciously not doing. A stage skipped on purpose costs nothing. A stage skipped without noticing is found in month five.

2 The three gates

These are the author's own, and they are prior to everything on the sheet. A project kicks start and materializes when all three are met, and not before.

1. **Experiment infrastructure implemented.** Not designed. Running.
2. **First evidence for the hypothesis in hand.** Something already observed that would be strange if the hypothesis were false.
3. **Algorithm sketch ready.** Concrete enough to hand over.

The third gate exists for a practical reason, and the reason is worth stating in the file so it does not get softened later. Students come from colleagues rather than from a group of one's own, and an idea that is not concrete enough will not attract them, nor will it be executable by them without the algorithm and the expected results handed over directly. The gate is not a standard of rigor. It is a condition for the work happening at all.

3 The sheet

Fifteen fields. Most are one line. The order matters, because each field constrains the next, and a field that cannot be written is the finding.

1. The question, written so it could be wrong. If no observation would settle it, it is a topic and not a question. Rewrite until it names something that could turn out either way.
2. The two fields it sits between. A question worth taking up usually sits where two literatures make different assumptions about the same object. Name both.
3. The challenge, in plain language. Before any terminology. If it cannot be said plainly it has not been understood yet.
4. The terminology, refined until it searches. Refine it until it matches the literature well enough to search and compare. This is also where the two fields in field 2 are checked for one term carried in two senses, which is itself a candidate project.
5. Why it matters. The scientific reason, not the funding reason.
6. The design move. The method, representation, or design decision used in response to the challenge. One sentence. If it takes three, there are probably two projects here.
7. The borrowed method, and which of three uses. There are three and they are not interchangeable. Integrate it as a device that makes the method work better. Take the algorithmic pattern and apply the same idea to your own model. Or take how they designed the experiments that showed their approach was better. Say which one, because the three cost very different amounts of work.
8. Contribution, and its evidence. Stated together or not at all.
9. The invariant. Not only what this project contributes, but what it helps establish that later work stands on: a problem class, a representational commitment, a methodological style, or an application domain in which later work stays credible.
10. Requirements, each with its verification. Write what the experiment must be able to show before writing the code that will show it. Every requirement on the way down has a verification on the way up. If you cannot say how a requirement will be verified, it is not a requirement, it is a hope.
11. Decision gates. Explicit points where you decide whether to continue, with the criteria fixed in advance. Research projects usually have no gates at all, which is why they end by exhaustion rather than by decision.
12. Technical risk, tracked. Likelihood times consequence, revisited, with the mitigation named beside each one.
13. Interfaces. Most integration failures sit at boundaries between components built by different people to different assumptions. True of spacecraft, true of a simulator and a policy learned somewhere else. List the boundaries and who owns each side.
14. Venue slice. Which claim goes to which venue, and the deadline.
15. Unsupported claims. The ones that need more reading or more data, listed rather than smoothed over. This field is never empty, and an empty one means the sheet was filled to look finished.

4 The four tests

Fields 1, 5, and the stopping rule below are the regulation layer. They are what people lack, as distinct from the operations they govern, and they are written as tests rather than as prompts for that reason.

1. **Could it be wrong?** Field 1. Name the observation that would settle it.

2. **Would the answer change something?** Name what would be done differently under each answer. A question qualifies the way a number does, by changing a decision.
3. **Is it reachable?** A question is reachable when an experiment you could run this year would move it. Otherwise it is a five year question and belongs on the vista rather than in an agenda.
4. **Has the space stopped producing candidates?** Stop when a full pass over both literatures adds nothing new. What remains then is choosing rather than searching.

5 Venues, as slices rather than as targets

One project usually carries more than one claim, and the claims are judged by different people. Slicing them is a design act and it belongs on the sheet rather than at submission time. The pattern already in use: the system paper to the human-computer interaction venue, the algorithmic contribution to the multiagent venue, the learning outcome to the education venue, and the cognitive claim to the cognitive science venue. Each gets the claim its reviewers can actually judge, and the earliest deadline is the one that has to clear all three gates at once.

6 Validation by convergence

Where the object being measured has no agreed definition, no single method establishes it. Validate by convergence across complementary methods of elicitation, each with a different blind spot. Agreement among independent methods confers confidence in an inferred object, and disagreement localizes where the inference is unreliable. The object is fixed not by one measurement but by the convergence of several. Five methods are already named for this: self-report, structured pairwise choices, process tracing through information boards, feature perturbation, and think-aloud protocols.

Field 10 is where this lands. A requirement whose verification is one measurement of an undefined object has not been verified.

7 What would kill it

The most useful thing a design sheet can carry is the condition under which the project is not worth doing, written before the work starts. The form that condition takes is a comparison against the simpler alternatives, stated as a rule rather than as a hope.

> The added component is supported only if it resolves a documented failure more reliably than every simpler alternative under the stated assumptions. If a simpler alternative supplies the same benefit, the evidence supports that architecture instead. If no alternative succeeds either, the result identifies an unresolved problem but does not by itself show that the added component is the missing piece.

The alternatives have to be listed by name in field 11 and each one has to be a gate. A comparison set assembled after the results are in is not a comparison set.

8 The macros

```
% One sheet. #1 the project's short name.
\newcommand{\pdsheet}[1]{\subsection*{#1}}

% The fields, kept as macros so a missing one is greppable.
\newcommand{\pdfield}[2]{\textbf{#1.} #2}

% A requirement and the verification that closes it. Never one without the other.
\newcommand{\pdreq}[2]{#1, verified by #2}

% A gate. #1 the decision point, #2 the criterion fixed in advance.
\newcommand{\pdgate}[2]{#1, continue only if #2}

% A risk. #1 the event, #2 likelihood times consequence, #3 the mitigation.
\newcommand{\pdrisk}[3]{#1 (#2), mitigated by #3}

% A stage from the systems engineering list that is being skipped on purpose.
\newcommand{\pdskip}[2]{#1, skipped because #2}

% A claim and the venue that can judge it.
\newcommand{\pdvenue}[3]{#1 to #2, deadline #3}
```

9 Filling it from a brainstorm

Question 34's note says to work from previous free-form chat histories. The order that turns that material into the sheet is seven steps, and it is already written in the gap plan.

1. Collect the informal intuition from notes, drafts, and meetings.
2. Identify the underlying challenge in plain language.
3. Refine the terminology until it matches the literature well enough to search and compare.
4. State the scientific reason the challenge matters.
5. Isolate the actual method, representation, or design move used in response.
6. State the concrete contribution and its evidence.
7. List the unsupported or weakly supported claims that require more reading or data.

Steps 1 through 7 fill fields 1 through 8 and field 15. Fields 9 through 14 are not in the brainstorm and never will be, because they are about execution rather than about the idea. Those are the ones that take the high energy hours.

One rule on the source material. The brainstorm is in your words, which is the reason it is worth keeping. Carry the sentences across rather than rewriting them. A sharpened argument built out of restated sentences is a different argument, and the restatement is where the sharpening stops.

10 Sources

- Fields 10 through 13, and the menu rule in section 1, are the five parts of the NASA Systems Engineering Handbook that transfer to a lab, as already written on the Project Management page of the website.
- The regulation argument, the four tests, and the stopping rule are from the Think page and from the problem finding note behind it.
- The seven step procedure in section 8, the invariant question in field 9, and the statement of what is missing in section 1 are from the gap plan section of the research statement.
- The three gates, the resource condition behind the third, the three uses of a borrowed method in field 7, and the venue slicing pattern are the author's own, from the project notes for the teaching game and for the world model failure work.
- The convergence argument and the five elicitation methods are from the validation section of the same project notes.
- The rule in section 7 is the decision rule already written for the agentic defender comparison, generalized here from causal integration to any added component.

Blog Stub Template

Openable in four minutes, publishable in one hour

1 What the stub is for

Question 36 states the problem and does not need restating: one or two hours a week are available for blogging, and without a stub to open, that hour goes to reading around while the thought stays suspended at the top of your mind and then vanishes.

The site already runs on stubs. Of the twenty-three pages under `content/overview/blogs`, eight carry a body. Thirteen are front matter and a title. Two sit in between: one holds a link and nothing else, one holds a single paragraph in italics. Four of the thirteen give `Stay tuned...` as their description.

So the practice exists. What it lacks is a shape and an order, so that the hour lands on writing rather than on deciding what the page is. Both are recoverable from the eight pages that are finished, because those eight agree with each other.

2 The shape, read off the finished pages

The six environment pages, the section front page above them, and the dictionary front page use the same ten parts in the same order. Not every page carries every part, and the ones that skip a part skip it from the bottom.

1. **Description**, in the front matter. One sentence, and it is the page's whole promise. The dictionary page uses it to state why the dictionary exists at all: "To collaborate is to communicate, to communicate is to cross the bridges of misunderstandings."
2. **Icon**, in the front matter. One name from the set GitBook offers.
3. **Title**. Two or three words. The name of the thing, nothing more.
4. **The gloss**. One bold line directly under the title saying what the thing is. All six environment pages have one.
5. **Figure and caption**. Five of the six have one. The caption is a verdict rather than a label: "Hard to search, quick to finish. Both at once." "Three axes, six environments. No single ordering." "Real hosts, simulated wire, real telemetry."
6. **Body**. Two to four paragraphs. The longest of the eight runs four.
7. **The verdict line**. Bold lead-in, then the use: `**Use it for**`, and where it applies, `**Don't use it for**`. Two of the six put a role in place of a use, naming the page's subject as the source or the far target of a transfer experiment.
8. **Publications**, under the italic lead-in `_Work of mine that runs on this environment._` Five of the six have one. The sixth replaces it with a `## Related` section pointing at the page where its question is taken up.

9. **Collaborators**, the same five pages, photo and affiliation per person.
10. **Footer**. `_Last updated: 2026-08_` on all eight.

Then one more part, which is not on the page: a line in `content/SUMMARY.md`. Without it the page is a file and not a page.

3 The stub itself, three fields

Three fields, two of them from the list above and one the standalone post supplies. The body is not among them.

> **Description.** The sentence you would give a colleague who asked what the post is about, before you have written any of it.
> **Title.** Two or three words.
> **The one sentence.** The thought that made you want the page. Written down in the shape it arrived in, including as a question.

The third field is the whole point and it is the one a stub usually skips. A page that has only a title is a filename with an intention attached, and a week later the intention no longer says what you meant. The interdisciplinary research post is the counter-example on the site: it has no body, and it holds an entire essay anyway, because its description names three things it will cover and its one italic paragraph asks the question the essay is about, starting "It's shocking, sometimes, to realize that there **are** solutions."

`Stay tuned...` is the description of last resort. It is honest, and it is also the sign that the thought was not written down while it was still there.

4 The skeleton

Copy this into a new `.md` file. Everything below the third field is optional at stub time and every line of it is a prompt rather than a requirement.

```
---
description: [one sentence, the promise of the page]
icon: [font-awesome-name]
---

# [Two or three words]

_[The one sentence. The thought, in the shape it arrived in.]_

**[The gloss: what this thing is, in one line.]**

<figure><img src="../../../.gitbook/assets/[name].gif"
  alt="[what is happening in the figure]"><figcaption><p>[Caption as a
```

```
  verdict, not a label.]</p></figcaption></figure>

[Body paragraph. What it is and how it works.]

[Body paragraph. What it costs, or what it gets wrong.]

**Use it for** [the case where it is the right choice].

**Don't use it for** [the case where it is not].

## Publications

_Work of mine that runs on this environment._

[table: badge | paper and venue | authors | link]

## Collaborators

[table: photo, name, affiliation, one cell per person]

_Last updated: 2026-08_
```

Three lines in that skeleton stay and the rest can go. The description, the one sentence, and the footer. A page without the footer looks abandoned rather than unfinished, which is a different and worse thing.

5 The fill order, and what fits in an hour

The order is not the order the parts appear in. Write the short fixed-cost parts first, because they are the ones that decide what the body has to say.

Part	**What it costs**	**When it gets written**
Description	one sentence	the minute the thought arrives
Title	seconds	same minute
The one sentence	one sentence	same minute
Gloss	one line	first hour
Verdict line	one or two lines	first hour
Caption	one or two short sentences	first hour
Body	two to four paragraphs	second hour
Figure	an asset to make	after the body, if at all
Tables	copied from a sibling page	last
`SUMMARY.md` line	one line	last

Write the verdict before the body. The verdict line is the sentence the post exists to deliver, and once it is on the page the body has something to reach. Writing it last means writing the body without knowing where it ends, which is how two paragraphs become six.

The caption is the second draft of the verdict. Every caption on the finished pages is two short sentences, or one sentence and a three-word correction after it. If the caption will not compress to that, the verdict is not sharp yet, and the body will not fix it.

Publish at the verdict line. A page with a description, a title, a gloss and a verdict is a page. It reads as short rather than as unfinished, and the body can arrive the following week.

6 Three kinds of page, three skeletons

Deciding which kind takes a few seconds and saves the rest of the hour.

The catalog entry. One member of a set the reader is comparing. The six environment pages. Gloss, figure, body, verdict, publications, collaborators. Its job is to be comparable to its siblings, so the parts stay in the same order and carry the same headings even where a page has nothing to put under one.

The section front page. Sits above a set and says what the set is for. The cyber environments page opens "Every claim about an autonomous cyber agent is a claim about the environment it was measured in." Then a figure, then the one paragraph naming the tension that makes the set worth reading as a set, then the linked list of children, then the footer. No tables. The children carry those.

The standalone post. No set, no siblings, no comparison. Description that carries the outline, one epigraph in italics, body. No tables, no verdict line. The interdisciplinary research post is this kind, and so is the PSRO post waiting to be written.

7 Registration, and the asset names

Three mechanical steps, all of them cheap and all of them easy to forget.

1. Add the page to `content/SUMMARY.md` at the indentation of its siblings. Nesting in that file is what produces the sidebar tree.
2. Put images in `content/.gitbook/assets/` using the prefixes already in use: `env-` for an environment animation, `badge-` for a venue or paper badge, `collab-` for a person. A page two levels below `content` reaches them at `../../../.gitbook/assets/`.
3. Copy the publication and collaborator tables from a sibling page rather than writing the markup. They are wide, they are fiddly, and the column widths are already tuned.

8 When a stub may stay a stub

For as long as you like. Thirteen of twenty-three have stayed stubs and the site is not worse for it, because a stub with a description is a promise the reader can read and decide about.

The one rule is the one the site already breaks in four places: a stub may not stay without a description. That is the field that costs a sentence and holds the thought, and it is the only part of the page that cannot be reconstructed later from the title.

9 Sources

- The ten parts, the order, the three page kinds, and every quoted line are read off the blogs tree on the site, `content/overview/blogs/`: the section front page and the six environment pages carry the body, the dictionary front page carries the description quoted in section 2, and the interdisciplinary research post carries the description-as-outline.
- The counts are of the twenty-three files in that directory as they stand in August 2026.
- The registration step and the asset prefixes are read from `SUMMARY.md` and from the assets directory beside it.
- The problem the template solves, and the hour budget it is built around, are stated in question 36.

Lexicon Template

One project, three tiers, four fields, two checks

1 What this fills

How to Build a Lexicon settles the method and the two sources under it, and the schema it uses is yours, from the Tech Fluency page. Neither is repeated here. This file is the skeleton you copy when a project's turn comes, plus the markdown for the page that carries the result.

One lexicon per project. A term used by more than one project goes in the shared file for its cluster and the project entries point at it. A term whose sense differs between two projects stays in both, and each entry says how the other project uses it.

2 The file, and the macros it already has

Lexicon files live in `_side/_lexicons` and share `_house.tex` with the census file. Four macros are defined there and nothing else needs defining.

```
\term{Stackelberg equilibrium}          % a headword
\says{the quoted sentence}{locator}     % your own sentence, with where it lives
\reviewer{B7}{their phrase}{locator}    % a reviewer's word for the same thing
\begin{clash}{one-line title}...\end{clash}   % one term, two senses
```

The opening of the file, unchanged from the two that exist:

```
\ifx\LexiconMaster\undefined
  \documentclass[11pt]{article}
  \input{_house}
  \begin{document}
\fi

\lxHeader{[Project name]}{[what the terms are drawn from]}

\tableofcontents
\clearpage
```

3 Step one, the census

Every term the project's own files use, with the sentence that uses it and a locator. No definitions at this stage, and no selection. The census is what makes the entries writable in your words rather than in someone else's, and it is the file you open to check whether the reading was real.

```
\term{[the term as you write it]}
\says{[your sentence, verbatim]}{[file, section]}
\says{[a second sentence, where a second file uses it]}{[file, section]}
```

The stopping rule is coverage. A complete pass over the project's files adds no new term, and every term found is assigned to a tier or marked ambiguous. Not a word count, not a term count, and not the point at which the list looks long enough.

4 Step two, the tier

Your three tiers, and what each one buys. The tier is decided before the entry is written, because it decides how much of the entry gets written.

Tier	The test	Fields written
Active	Must define and use fluently, on the spot	All four
Working	Must recognize precisely; lookup is fine	Definition, Distinguishes it from
Peripheral	Know approximately where it belongs and retrieve when necessary	Definition, and a pointer to where it belongs

Only the Active tier reaches the website page. Working and Peripheral stay in the compiled file, which the page links to.

5 Step three, the entry

Four fields, in your order, on your Stackelberg example.

Stackelberg equilibrium
Definition: Leader commits; follower observes and best-responds.
Distinguishes it from: Nash: simultaneous strategic choice, no commitment advantage.
Operational test: Can one player commit before the other's decision?
In my project: Defender commits to a strategy; attacker responds.

5.1 Definition

One sentence. Taken from your manuscript where your manuscript defines it. Taken from the field's own source where it does not, with the source named. Written fresh only where neither has it, and said to be so.

Where the formal definition and your working understanding differ, both go in and each is labeled. The difference is the thing worth having, and a lexicon that records only the formal one hides it.

5.2 Distinguishes it from

The nearest neighbor, and the one property that separates the two. One neighbor per entry unless a second is genuinely as close.

This is the field that does the detecting. A person can hold a definition and still fail at the first unfamiliar case, and level one of your fluency scale will not catch it. Naming what the term is not is what catches it.

5.3 Operational test

A question with a yes or no answer that decides whether the term applies to a case in front of you. If the answer requires a paragraph, the test is not written yet.

Two failures to watch. A test that restates the definition tests nothing. A test that can only be answered by someone who already knows the answer is a definition wearing a question mark.

5.4 In my project

Where the term appears in this project's own work, with the locator. Concrete enough that the sentence can be found again: file and section, not file alone.

6 The clash, for a term whose two uses disagree

A term whose *In my project* field cites two files that use it differently does not get resolved into one entry. It gets a clash box, and the clash box is the most valuable thing in the file.

```
\begin{clash}{[one line, stating the disagreement]}
[Source one]: ``[the sentence]'', [locator].

[Source two]: ``[the sentence]'', [locator].

[One or two sentences saying what the disagreement is. Not which side
is right, unless you know.]
\end{clash}
```

The shortest one in the FOE-Dreamer census is three lines: a term appears in the deployment notes, is not one of the five defender actions, and appears nowhere else in the paper. That is a complete clash. It needs no resolution to be worth writing down.

7 Step four, the two checks

Both are cheap and both catch a wrong entry rather than a missing one.

Run the discrimination test in both directions. If the entry for one term names a second term as its neighbor, the second term's entry names the first. A one-way distinction usually means one of the two definitions is wrong.

Look for two locators that disagree. Any entry whose *In my project* field cites two files using the term differently becomes a clash box. These are the entries a reviewer trips on, and finding them is most of what the lexicon is for.

8 Placement, for a term that crosses communities

A term that cybersecurity, AI, and human factors all use is placed on the five dimensions from the dictionary page rather than defined three times. The row is the placement.

Dimension	**The range it runs over**
Object	actor, behavior, attack possibilities, environment
Epistemic operation	describe, infer, predict, simulate or enact
Specificity	generic adversary, adversary class, particular actor
Purpose	strategic response, risk analysis, defensive evaluation
Fidelity	abstract representation, executable reproduction

Two lines go under the table, because the table cannot carry them.

Salience. Which dimension decides the term. A concept is a set of regions plus weights on the dimensions, and a table of regions alone has dropped the weights.

Dimensions that move together. Where specificity rises with fidelity, or any other pair travels as one, say so in a line. Boxes on independent axes cannot express it, and nothing in the table rules out the combination that does not exist.

Then one check, run on the dimension rather than on the term. If two terms sit at two points on a dimension and a term at an intermediate point would be an instance of neither, that dimension is not carrying the distinction you think it is. Drop it or split it.

9 The website page

Each project's `lexicon.md` under `content/overview/3-year-agenda/` is currently front matter and a heading. Filled, it carries the Active tier and links to the compiled file, the way `resources/templates.md` does for the beamer theme.

```
---
description: [one sentence: what these terms are for]
icon: spell-check
---

# Lexicon

[One paragraph: which files the terms were taken from, and what the
tiers mean here.]
```

```
<table><thead><tr><th width="220">Term</th><th>Definition</th>
<th>Distinguishes it from</th></tr></thead><tbody>
[one row per Active term]
</tbody></table>

**Operational tests and project locations** are in the compiled file.
[**Download the lexicon (.pdf)**](...)

_Last updated: 2026-08_
```

Three columns on the page, four fields in the file. The operational test and the project location are the two that need a locator to be useful, and a locator is worth nothing to a reader who does not have the repository.

10 Filling it

The order is the four steps, and only step three writes. Census, tier, entries, checks. A project whose census is not finished does not get entries started, because an entry written from memory is an entry written in the field's words rather than yours, and the whole apparatus exists to prevent that.

11 Sources

- The three fluency levels, the Active, Working and Peripheral tiers, the four fields and the Stackelberg example are yours, from the Tech Fluency page, `content/overview-2/tech-fluency-from-concept-image-to-concept-defintions.md`. The example is reproduced with its internal dash written as a colon.
- The five dimensions and the ranges they run over are yours, from the Cyber-Human-AI Dictionary front page.
- The justification for the four fields, the salience and correlation lines, and the convexity check are in `_side/_lexicons/lexicon_method.tex`, which cites Tall and Vinner 1981 and Gärdenfors for them.
- The macros, the census shape and the clash shape are from `_side/_lexicons/_house.tex` and `_side/_lexicons/your_terms_foedreamer.tex`.